

- brain 78, **86–97**
 ageing 285
 anatomy 86–7
 autonomic nervous system 108
 blood supply 86
 cranial nerves **102–3**
 disorders **124–6**
 growth and development 277
 infections **126**
 information processing **20**
 memories, thoughts, and emotions **110–11**
 nerve damage 83
 primitive brain **94–5**
 protection 88
 scanning 13
 sense of taste and smell 113
 sense of touch 115
 sleep cycles 97, 138
 stroke 157
 structures 86–7, **90–1**
 tissues 36
 vision 120, 122
 brainstem 87, 90, 93, 96
 breastbone 40, 52
 breast-feeding 273
 breasts 259
 disorders 296
 milk production 135
 in pregnancy 265
 in puberty 282
 breathing **166–70**, 217, 272
 see also respiratory system
 bronchi 162–5
 bronchioles 162–5, 167, 173
 bronchitis 173, 175
- C**
 caecum 224, 225, 232
 calcium 42, 58, 137
 cancer **305**
 breast 296
 colorectal 239
 lung 174
 prostate 294
 skin 12, 187
 testicular 295
Candida albicans 207, 209, 299
 canine teeth 216, 278
 capillaries 146, 149
 lymphatic system 192, 193
 in skin 179
 vasodilation 201
 carbohydrates 229, 232, 233
 carbon dioxide: ageing 284
 in blood 21
 cardiovascular system 146
 gas exchange 164, **166–7**
 cardiac muscle 67
 angina 157
 heart attack 156
 heartbeat 155
 cardiac skeleton 150
 cardiovascular system 15, **144–59**
 anatomy **146–7**
 blood and blood vessels **148–9**
 disorders **156–9**
 heart **150–1**
 heartbeat **154–5**
 homeostasis 23
 in newborn baby 275
 carpal bones 40, 44, 54
 carpal tunnel syndrome 75
 cartilage 43
 bone development 42, 276
 cells 36
 ears 116
 intervertebral discs 50
 joints 44, 46, 47
 larynx 171
 in newborn baby 272
 osteoarthritis 60
 ribcage 52
 torn cartilage 59
 catabolism 233
 cataracts, eyes 285
 cells 10, **24–37**
 ageing 284
 anatomy 27
 blood 148, 194
 bone 42
 cancer 305
 cell membrane 26, 27, 28
 cellular respiration 166
 development of embryo 260–3
 division 289
 DNA **30–1**, 286
 egg cells 256
 epithelial tissues 184
 genetic control 35
 growth, renewal, and repair 233
 inflammatory response 198–201
 muscle 70
 neurons 80–1
 sex cells 289
 skin renewal 178
 sperm 252
 tissues 36–7
 types **27**
 central nervous system (CNS) 78
 centriole, cells 26
 cerebellum 86, 87, 90–1, 109
 cerebral cortex 90, 92, 109, 113
 cerebrospinal fluid (CSF) 83
 brain 88, 89, 91
 spinal cord 98, 100
 cerebrovascular disorders **124**
 cerebrum 86, 87, 90–1, 92
 cervical vertebrae 41, 44, 51
 cervix 257, 264
 birth 270
 fertility problems 300
 menopause 285
 placenta praevia 302
 preparation for birth 267, 268–9
 chemotaxis 201
 chest: breathing 168–70
 muscles 64, 66
 nervous system 78
 children: growth and development **276–9**
 puberty **280–3**
 chlamydia 298, 299
 chloride, digestive system 232
 choking 217
 cholesterol: atherosclerosis 157
 in blood 148
 in colon 232
 gallstones 238
 chondrocytes 36, 43
 chromosomes: cell division 289
 disorders **304**
 DNA 30–1, 33
 fertilization of egg 289
 and gender 291
 genome **34–5**, 286–7
 patterns of inheritance 290–1
 sex cells 289
 chronic obstructive pulmonary disease (COPD) 175
 chyme 219, 224, 228, 229
 cilia 28, 184, 198, 260
 circle of Willis 86
 circulatory system see cardiovascular system
 cirrhosis, liver 236, 237
 clavicle 40, 45
 clitoris 257, 258
 clots see blood clots
 cocci 204
 coccyx 41, 50, 51, 53
 cochlea 116–18, 128, 285
 cold, common 172
 cold sores 188
 collagen 44, 55, 284
 collarbone 40
 colon 213, **224–7**, 232, 239
 colostrum 273
 colour: colour blindness 293
 eyes 33, 292
 skin pigmentation 187
 columnar cells 184
 communication: facial expressions 68
 see also speech
 compact bone 42, 43
 complement system, immune system 196–7
 computerized tomography (CT) 13
 conception 260, 289
 cone cells, retina 120
 conjunctiva, eye 121
 connective tissue 36, 37
 ageing 284
 cartilage 43
 joints 46
 tendons 72
 contractions, labour 267–70
 cornea 120, 121, 129, 185
 coronary arteries 150, 153
 coronary heart disease **156**
 coronary veins 150
 corpus callosum 87, 92
 corpus luteum 258, 283
 cortex, brain 110, 111
 coughing 171, 217
 cranial nerves **102–3**, 113
 cranium 48, 277
 cuboidal cells 185
 cystic fibrosis 304
 cystitis 248
 cysts 189, 296
 cytoplasm 21, 26, 29
 cytosine 30, 31, 287
 cytoskeleton, cells 26
- D**
 deafness 128, 285
 defecation 227
 defences: immune system **194–209**
 skin and hair **186–7**
 delivery, birth **270–1**